

Database Project Report Template

Due _____ **Date:** _____ [Week _____ 5-8.]
Project ID & Title: University Research & Lab Asset Tracker

A. Project Identity

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- **Project Title :** University Research & Lab Asset Tracker

B. Report structure

1. Introduction & Project Scope

1.1 System Objective

1.1.1 Problem Statement

Universities operate multiple research labs serving faculty and students, using high-value equipment and chemicals while also receiving funding from multiple research grants provided by different funding organizations. Currently, the management of lab assets, chemical inventory, and lab access rights is often carried out manually using paper records and separate spreadsheets, leading to the following problems:

- It is difficult to control who is accessing hazardous chemicals or restricted areas, creating occupational safety risks.
- Equipment and chemicals are not associated with a specific lab or grant, making them difficult to trace during audits.
- There is no mandatory constraint ensuring that only personnel with appropriate safety certifications are allowed to access hazardous materials.
- There is insufficient data linkage among personnel (faculty/students), research grants, and the laboratory assets they use.

The objective of the project is to design and build a relational database named "University Research & Lab Asset Tracker" for the centralized management of: (1) research grants, (2) laboratory equipment, (3) chemical inventory, and (4) access logs for faculty and students entering or leaving laboratories. The most important integrity constraint is that only personnel with valid, unexpired certifications may access materials or areas with the corresponding hazard level. This constraint will be enforced at the database layer through an SQL Trigger.

1.1.2 Scope

- In scope: management of departments, personnel (faculty/students), safety certifications, laboratories, equipment, chemicals, research grants, and access logs.
- Out of scope: the university's full financial accounting system, academic administration (course registration), and the complete HR system for university employees.

1.1.3 Functional Requirements

- FR1: The system allows an Administrator (Admin) to register new laboratory equipment and assign the equipment to a specific laboratory.
- FR2: The system allows Faculty members to create and manage research grant records, including the funding organization name, grant amount, and project duration.
- FR3: The system records every personnel access event at a laboratory, including the timestamp and access type (entry/exit).
- FR4: The system must validate a personnel member's safety certification before allowing access to chemicals or laboratories classified as hazardous. This rule is enforced by an SQL Trigger, not only at the application layer.
- FR5: The system tracks chemical inventory, including stock quantity, storage location, and hazard classification (hazard class).
- FR6: The system allows multiple personnel members (faculty/students) to be assigned to a grant with specific roles (PI, Co-PI, Research Assistant).
- FR7: The system supports report queries, including lists of unauthorized access attempts, equipment by laboratory, and chemicals with related certifications that are nearing expiration.

1.1.4 Non-Functional Requirements

- NFR1 (Security): Certification data and personnel identification information must be subject to strict access control; encryption at rest is recommended for sensitive fields.
- NFR2 (Performance): Access-log lookup queries used for auditing must return results in under 200 ms with up to 100,000 records.
- NFR3 (Availability): The system should be available 24/7 because laboratories may be accessed outside normal working hours.
- NFR4 (Data Integrity): The rule that "only personnel with valid certifications may access hazardous materials" must be enforced at the database layer (trigger/constraint) and must not depend entirely on the application layer.

1.2. Business Rules

1.2.1 Personnel & Certification

- BR1: Each Personnel has a unique person_id. A Personnel may be specialized as either Faculty or Student. The specialization is disjoint and partial; therefore, a Personnel member may belong to neither subtype.
- BR2: Each Faculty member belongs to exactly one Department; a Department may have many Faculty members.
- BR3: Each Student may have at most one Faculty member as an advisor; a Faculty member may advise many Students.
- BR4: A Personnel member may be issued zero or more Certifications; a Certification may be issued to many Personnel members (an N:N relationship through PERSON_CERTIFICATION), and each issuance has its own issue_date and expiry_date.
- BR5: A Certification is valid only when the current date falls within [issue_date, expiry_date]; expired certifications are not considered valid when determining access permission.

1.2.2 Laboratories, Equipment & Chemicals

- BR6: Each Equipment item belongs to exactly one Lab; a Lab may contain zero or more Equipment items.
- BR7: Each Chemical is stored in exactly one Lab; a Lab may store zero or more Chemicals.
- BR8: Each Lab and each Chemical has a hazard_level/hazard_class. If hazard_level != 'None', only Personnel with a valid Certification at the corresponding hazard level or a higher level may access it (this constraint is enforced by an SQL Trigger BEFORE INSERT on ACCESS_LOG).

1.2.3 Research Grants

- BR9: Each Research Grant has exactly one Principal Investigator (PI), and the PI must be a Faculty member (not a Student).
- BR10: A Faculty member may be the PI of multiple Grants.
- BR11: A Grant may have many participating Personnel members with different roles (PI, Co-PI, Research Assistant) through the GRANT_MEMBER associative table; a Personnel member may participate in multiple Grants.
- BR12: The Grant amount must be > 0, and end_date must be later than start_date.

1.2.4 Access Log

- BR13: Each Access Log record must reference an existing valid Personnel member and Lab; it may optionally reference a specific Equipment item and/or Chemical if the event relates to that item or substance.
- BR14: Each access event has an access_type of 'Entry' or 'Exit' and an authorized_flag automatically determined by the trigger based on the Personnel member's valid Certification relative to the hazard_level of the related Lab/Chemical.
- BR15: The system does NOT allow an Access Log record to be inserted (INSERT) with authorized_flag = TRUE if the Personnel member does not have the corresponding valid Certification. This is the central DB Challenge of the project and will be implemented using an SQL Trigger in Phase 3.

2. Database Design

2.1 Conceptual Model

2.1.1 Comprehensive ER/EER Diagram



Entity	Primary Key (PK)	Purpose
DEPARTMENT	dept_id	Department/unit responsible for managing faculty members.
PERSONNEL	person_id	Common supertype for all personnel (faculty, students, etc.).
FACULTY	person_id (FK)	Subtype: faculty member; may serve as a grant PI and advise students.
STUDENT	person_id (FK)	Subtype: student; may have a Faculty member as an advisor.
CERTIFICATION	cert_id	Safety certification type / authorized hazard-access level.
PERSON_CERTIFICATION	person_id, cert_id	N:N associative table recording certifications issued to each person and their validity.
LAB	lab_id	Laboratory with its own hazard level.

EQUIPMENT	equipment_id	Equipment assigned to a specific Lab.
CHEMICAL	chemical_id	Chemical stored in a Lab, with a hazard_class.
RESEARCH_GRANT	grant_id	Research grant with one PI (Faculty).
GRANT_MEMBER	grant_id, person_id	N:N associative table between a Grant and participating Personnel, including roles.
ACCESS_LOG	log_id	Lab entry/exit log with optional Equipment/Chemical references and an authorized_flag determined by a trigger.

2.1.3 Relationship Summary

Relationship	Cardinality (min,max)	Related Business Rule
DEPARTMENT - FACULTY	1,1 - 0,N	BR2
FACULTY - STUDENT (advises)	0,1 - 0,N	BR3
PERSONNEL - PERSON_CERTIFICATION - CERTIFICATION	1,1 -(0,N)-(1,N)- 1,1	BR4, BR5
LAB - EQUIPMENT	1,1 - 0,N	BR6
LAB - CHEMICAL	1,1 - 0,N	BR7
FACULTY - RESEARCH_GRANT (is PI of)	1,1 - 1,N	BR9, BR10
RESEARCH_GRANT - GRANT_MEMBER - PERSONNEL	1,1 -(1,N)-(0,N)- 0,1	BR11
PERSONNEL - ACCESS_LOG	1,1 - 0,N	BR13
LAB - ACCESS_LOG	1,1 - 0,N	BR13
EQUIPMENT - ACCESS_LOG (optional)	0,1 - 0,N	BR13
CHEMICAL - ACCESS_LOG (optional)	0,1 - 0,N	BR13
PERSONNEL -(d)- FACULTY / STUDENT	EER disjoint, partial specialization	BR1

2.1.4 Design Assumptions

- A Personnel member cannot be both Faculty and Student at the same time (disjoint specialization).
- A Grant has only one PI at a time; if the PI changes, pi_id is updated (the PI change history is not stored in Phase 1).
- Each Access Log record is associated with at most one Equipment item OR one Chemical (neither is mandatory), supporting detailed traceability in accordance with BR13.
- The hazard level (hazard_level/hazard_class) uses a consistent categorical scale across Lab, Chemical, and Certification (for example: None, Low, Medium, High) so that the trigger in Phase 3 can compare them directly.

2.2 Logical Schema Mapping

Mapping Rules Applied:

Rule 1 — Strong Entity: Each strong entity maps to a separate relation table. The entity's key becomes the table's PK.

Rule 2 — Single-valued Attributes: Each single-valued attribute becomes a column in the table.

Rule 3 — Binary 1:N Relationship: The PK of the '1' side is added to the 'N' side table as a FK. Example:
EQUIPMENT.lab_id → LAB.lab_id.

Rule 4 — Binary N:N Relationship: A new association table is created with FKs from both sides. The composite PK consists of those FKs. Applied to PERSON_CERTIFICATION and GRANT_MEMBER.

Rule 5 — EER Specialization (Method 8A): For DISJOINT PARTIAL specialization (PERSONNEL → FACULTY/STUDENT): retain the supertype table PERSONNEL and create separate subtype tables FACULTY and STUDENT. Each subtype's PK is a FK referencing PERSONNEL with ON DELETE CASCADE. Chosen because: (a) avoids excessive NULL values, (b) supports partial specialization (personnel in neither subtype remain valid), (c) enables efficient subtype-specific queries.

Rule 6 — Optional FK (Nullable): Optional relationships are represented by nullable FKs. Example:
ACCESS_LOG.equipment_id and ACCESS_LOG.chemical_id may be NULL when the event is not linked to a specific item.

Full Relational Schema (Text Notation):

DEPARTMENT (dept_id [PK], dept_name, building)

PERSONNEL (person_id [PK], full_name, email, phone, person_type)

FACULTY (person_id [PK, FK*→PERSONNEL], dept_id [FK*→DEPARTMENT, null], tenure_status, office_location)

STUDENT (person_id [PK, FK*→PERSONNEL], program, enrollment_year, advisor_id [FK*→FACULTY, null])

CERTIFICATION (cert_id [PK], cert_name, hazard_level, description)

PERSON_CERTIFICATION (person_id [PK, FK*→PERSONNEL], cert_id [PK, FK*→CERTIFICATION], issue_date, expiry_date, status)

LAB (lab_id [PK], lab_name, building, room_number, hazard_level)

EQUIPMENT (equipment_id [PK], name, model, serial_number, purchase_date, status, lab_id [FK*→LAB])

CHEMICAL (chemical_id [PK], name, cas_number, hazard_class, quantity, unit, storage_location, lab_id [FK*→LAB])

RESEARCH_GRANT (grant_id [PK], title, funding_agency, amount, start_date, end_date, pi_id [FK*→FACULTY])

GRANT_MEMBER (grant_id [PK, FK*→RESEARCH_GRANT], person_id [PK, FK*→PERSONNEL], role_on_grant)

ACCESS_LOG (log_id [PK, AUTO_INCREMENT], person_id [FK*→PERSONNEL], lab_id [FK*→LAB], equipment_id [FK*→EQUIPMENT, null], chemical_id [FK*→CHEMICAL, null], access_timestamp, access_type, authorized_flag)

Foreign Key Summary:

Child Table	FK Column	References (Parent)	ON DELETE	Business Rule
FACULTY	dept_id	DEPARTMENT(dept_id)	SET NULL	BR2
FACULTY	person_id	PERSONNEL(person_id)	CASCADE	BR1
STUDENT	person_id	PERSONNEL(person_id)	CASCADE	BR1
STUDENT	advisor_id	FACULTY(person_id)	SET NULL	BR3

PERSON_CERTIFICATION	person_id	PERSONNEL(person_id)	CASCADE	BR4
PERSON_CERTIFICATION	cert_id	CERTIFICATION(cert_id)	RESTRICT	BR4
EQUIPMENT	lab_id	LAB(lab_id)	RESTRICT	BR6
CHEMICAL	lab_id	LAB(lab_id)	RESTRICT	BR7
RESEARCH_GRANT	pi_id	FACULTY(person_id)	RESTRICT	BR9
GRANT_MEMBER	grant_id	RESEARCH_GRANT(grant_id)	CASCADE	BR11
GRANT_MEMBER	person_id	PERSONNEL(person_id)	CASCADE	BR11
ACCESS_LOG	person_id	PERSONNEL(person_id)	RESTRICT	BR13
ACCESS_LOG	lab_id	LAB(lab_id)	RESTRICT	BR13
ACCESS_LOG	equipment_id	EQUIPMENT(equipment_id)	SET NULL	BR13
ACCESS_LOG	chemical_id	CHEMICAL(chemical_id)	SET NULL	BR13

2.3 Normalization Verification

Normalization eliminates data redundancy and ensures integrity. We verify each relation from 1NF through BCNF using Functional Dependency (FD) analysis.

- **1NF:** All attributes are atomic (no repeating groups, no multi-valued attributes).
- **2NF:** Achieves 1NF AND no partial dependency: every non-key attribute depends on the WHOLE PK (only relevant for composite PKs).
- **3NF:** Achieves 2NF AND no transitive dependency: for every FD $X \rightarrow Y$, X must be a superkey or Y must be a prime attribute.
- **BCNF:** Stricter than 3NF: for every FD $X \rightarrow Y$, X must be a superkey. Eliminates all redundancy based on FDs.

Table: DEPARTMENT

Primary Key / Candidate Keys	PK: dept_id Candidate Keys: dept_id
Functional Dependencies (FDs)	dept_id $\rightarrow$ dept_name, building
Normalization Proof	1NF: All attributes atomic. 2NF: Single-attr PK, no partial deps. 3NF: No transitive deps. BCNF: Only determinant is dept_id (superkey). Conclusion: Achieves BCNF.
Conclusion	DEPARTMENT achieves BCNF.

Table: PERSONNEL

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id, email
Functional Dependencies (FDs)	person_id $\rightarrow$ full_name, email, phone, person_type email $\rightarrow$ person_id, full_name, phone, person_type
Normalization Proof	1NF: All attributes atomic. 2NF: Single-attr PK. 3NF: No transitive deps. BCNF: Both determinants (person_id, email) are candidate keys = superkeys. Conclusion: Achieves BCNF.
Conclusion	PERSONNEL achieves BCNF.

Table: FACULTY

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id
Functional Dependencies (FDs)	person_id -> dept_id, tenure_status, office_location
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: dept_id is FK, does not cause transitive dependency in FACULTY table as department name is not stored here. Conclusion: Achieves BCNF.
Conclusion	FACULTY achieves BCNF.

Table: STUDENT

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id
Functional Dependencies (FDs)	person_id -> program, enrollment_year, advisor_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: advisor_id is FK, does not cause transitive dependency. Conclusion: Achieves BCNF.
Conclusion	STUDENT achieves BCNF.

Table: CERTIFICATION

Primary Key / Candidate Keys	PK: cert_id Candidate Keys: cert_id, cert_name
Functional Dependencies (FDs)	cert_id -> cert_name, hazard_level, description cert_name -> cert_id, hazard_level, description
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: Both determinants (cert_id, cert_name) are candidate keys. Conclusion: Achieves BCNF.
Conclusion	CERTIFICATION achieves BCNF.

Table: PERSON_CERTIFICATION

Primary Key / Candidate Keys	PK: (person_id, cert_id) Candidate Keys: (person_id, cert_id)
Functional Dependencies (FDs)	(person_id, cert_id) -> issue_date, expiry_date, status
Normalization Proof	1NF: Atomic. 2NF: Single person_id or cert_id does not determine issue_date / expiry_date - no partial dependency. 3NF & BCNF: Only one FD, left side is composite PK (superkey). Conclusion: Achieves BCNF.
Conclusion	PERSON_CERTIFICATION achieves BCNF.

Table: LAB

Primary Key / Candidate Keys	PK: lab_id Candidate Keys: lab_id, lab_name, (building, room_number)
Functional Dependencies (FDs)	lab_id -> lab_name, building, room_number, hazard_level (building, room_number) -> lab_id, lab_name, hazard_level
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: Every determinant is a candidate key (superkey). Conclusion: Achieves BCNF.
Conclusion	LAB achieves BCNF.

Table: EQUIPMENT

Primary Key / Candidate Keys	PK: equipment_id Candidate Keys: equipment_id, serial_number (khi NOT NULL)
Functional Dependencies (FDs)	equipment_id -> name, model, serial_number, purchase_date, status, lab_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: No transitive dependency. lab_id

	is FK but lab name/details are not stored here. Conclusion: Achieves BCNF.
Conclusion	EQUIPMENT achieves BCNF.

Table: CHEMICAL

Primary Key / Candidate Keys	PK: chemical_id Candidate Keys: chemical_id, cas_number (khi NOT NULL)
Functional Dependencies (FDs)	chemical_id -> name, cas_number, hazard_class, quantity, unit, storage_location, lab_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: No transitive dependency. lab_id is FK. Conclusion: Achieves BCNF.
Conclusion	CHEMICAL achieves BCNF.

Table: RESEARCH_GRANT

Primary Key / Candidate Keys	PK: grant_id Candidate Keys: grant_id
Functional Dependencies (FDs)	grant_id -> title, funding_agency, amount, start_date, end_date, pi_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: pi_id is FK, PI name is not stored here. No transitive dependency. Conclusion: Achieves BCNF.
Conclusion	RESEARCH_GRANT achieves BCNF.

Table: GRANT_MEMBER

Primary Key / Candidate Keys	PK: (grant_id, person_id) Candidate Keys: (grant_id, person_id)
Functional Dependencies (FDs)	(grant_id, person_id) -> role_on_grant
Normalization Proof	1NF: Atomic. 2NF: Single grant_id or single person_id does not determine role_on_grant - no partial dependency. 3NF & BCNF: Only one FD, left side is composite PK. Conclusion: Achieves BCNF.
Conclusion	GRANT_MEMBER achieves BCNF.

Table: ACCESS_LOG

Primary Key / Candidate Keys	PK: log_id Candidate Keys: log_id
Functional Dependencies (FDs)	log_id -> person_id, lab_id, equipment_id, chemical_id, access_timestamp, access_type, authorized_flag
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK (AUTO_INCREMENT). 3NF & BCNF: authorized_flag is determined by trigger based on business logic, not an FD in schema. No transitive dependency. Conclusion: Achieves BCNF.
Conclusion	ACCESS_LOG achieves BCNF.

Normalization Summary:

Table	1NF	2NF	3NF	BCNF	PK Type
DEPARTMENT	Pass	Pass	Pass	Pass	Single
PERSONNEL	Pass	Pass	Pass	Pass	Single
FACULTY	Pass	Pass	Pass	Pass	Single
STUDENT	Pass	Pass	Pass	Pass	Single
CERTIFICATION	Pass	Pass	Pass	Pass	Single
PERSON_CERTIFICATION	Pass	Pass	Pass	Pass	Composite

LAB	Pass	Pass	Pass	Pass	Single
EQUIPMENT	Pass	Pass	Pass	Pass	Single
CHEMICAL	Pass	Pass	Pass	Pass	Single
RESEARCH_ GRANT	Pass	Pass	Pass	Pass	Single
GRANT_ME MBER	Pass	Pass	Pass	Pass	Composite
ACCESS_LO G	Pass	Pass	Pass	Pass	Single (AUTO_INCR EMENT)